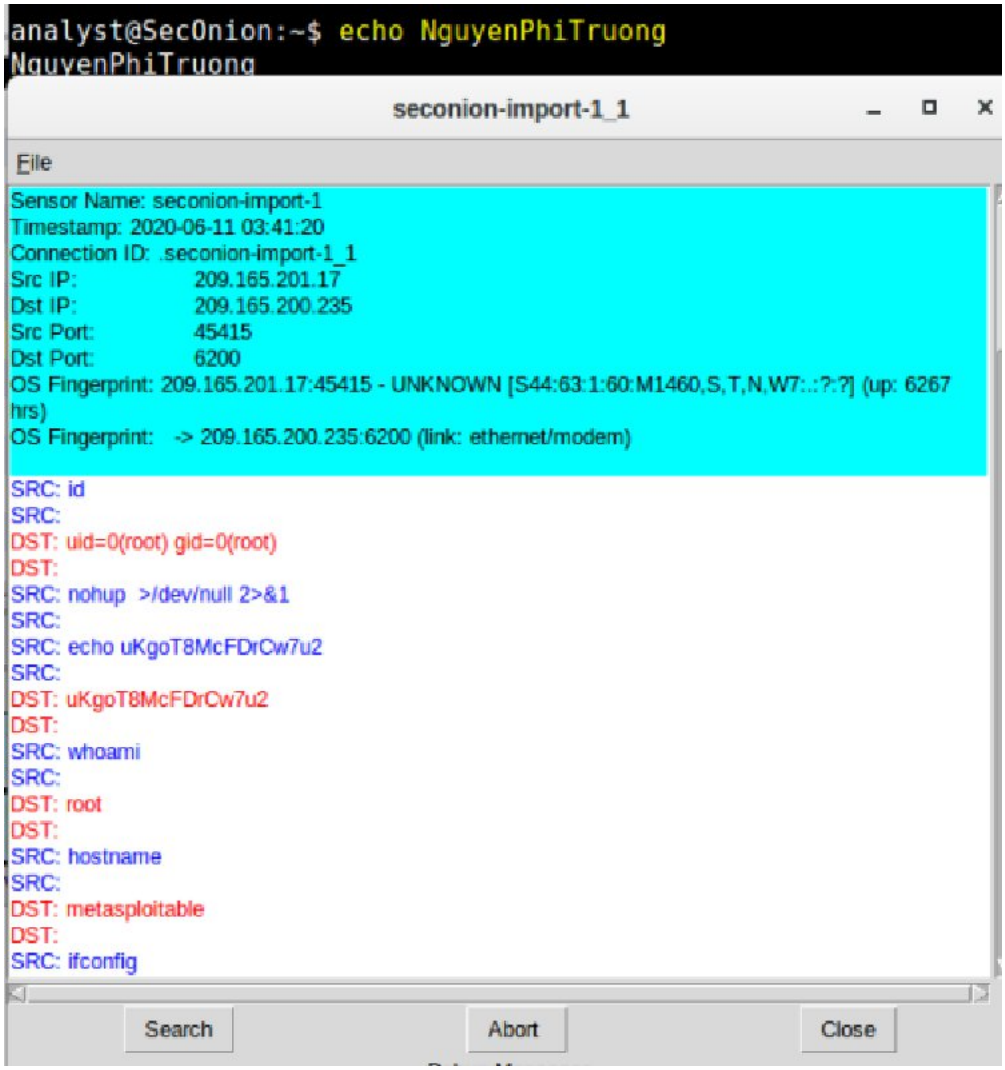
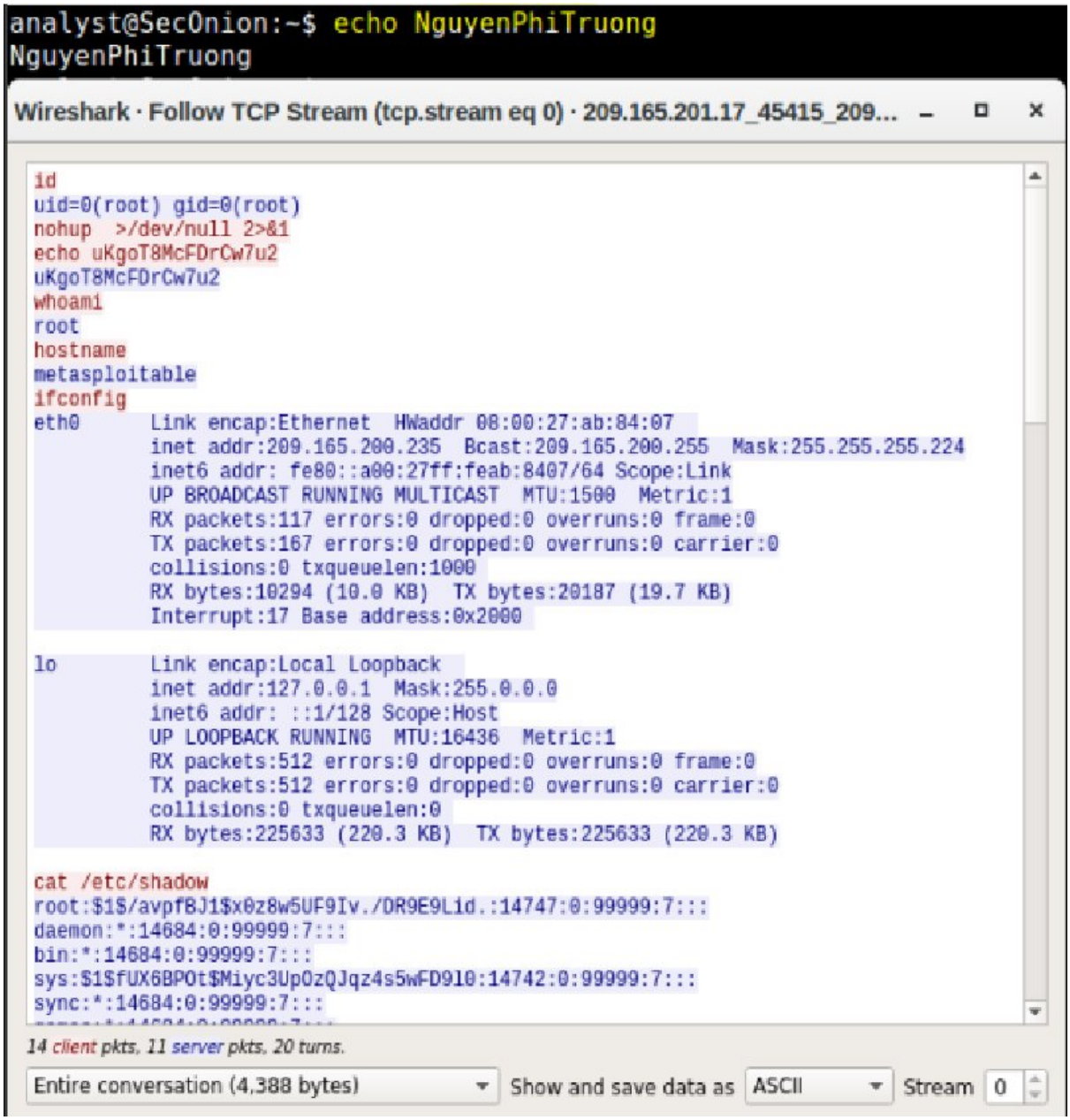
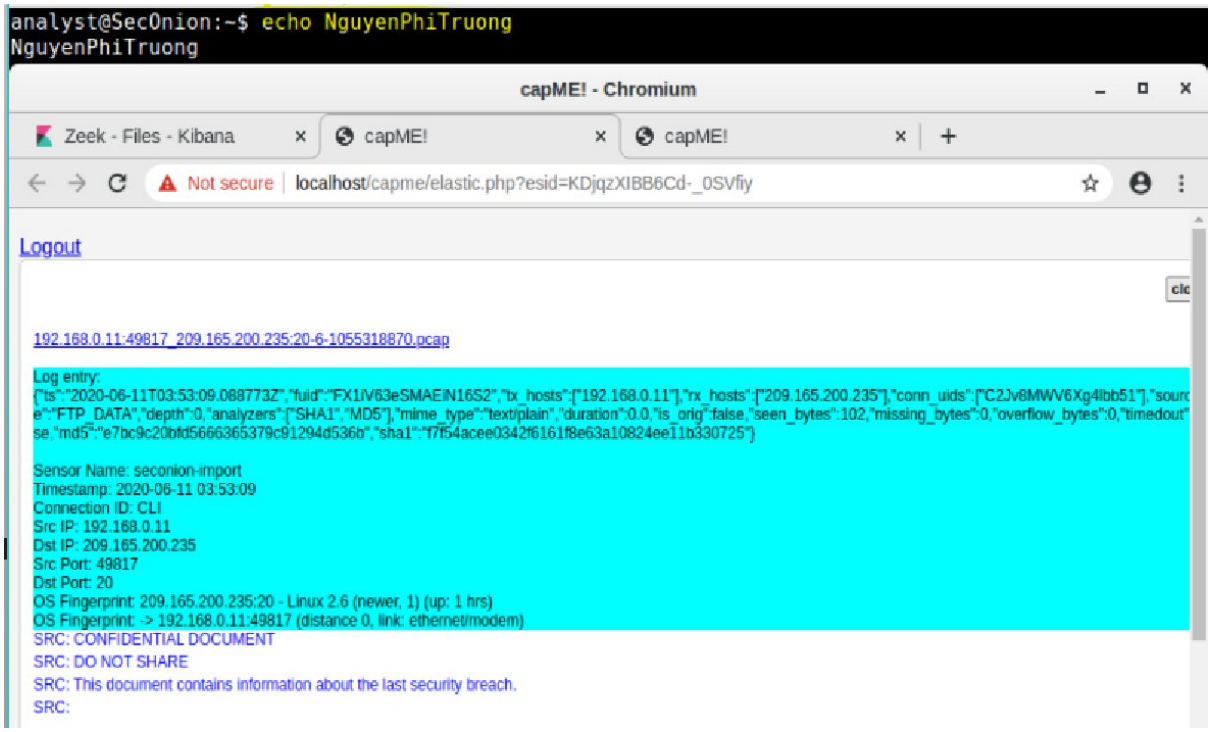


The window below shows the transaction log between the attacker (SRC - blue text) and the target machine (DST - red text) over port 6200. The attacker executes commands like id and whoami. The target responds with root, confirming that the hacker has successfully gained the highest administrative privileges on the system. The hostname command reveals the target is a metasploitable machine.



This Wireshark window displays the raw network conversation assembled from packets. The red text is the attacker's input, and the blue text is the target's output. The attacker runs reconnaissance commands like whoami, hostname, and ifconfig. Most importantly, at the bottom, the attacker executes cat /etc/shadow. This indicates they are viewing and stealing the system's hashed password file to compromise user accounts.



The cyan highlight section displays the technical details of the data flow between the attacker's machine (192.168.0.11) and the target server (209.165.200.235) over the FTP Data port (port 20). The purple/blue text at the bottom is the ultimate piece of evidence. It reveals the exact plain-text payload of the exfiltrated confidential.txt file: *"CONFIDENTIAL DOCUMENT / DO NOT SHARE / This document contains information about the last security breach."*.

1. ***What is Sguil and what is its main purpose in network security monitoring?***
   Sguil is a graphical tcl/tk interface designed for network security monitoring (NSM). Its main purpose is to provide security analysts with a centralized console to view security alerts, analyze network session data, and easily pivot to other investigative tools like Wireshark or Kibana to analyze potential intrusions.
2. ***With all the information has gathered so far, what is your recommendation for stopping further unauthorized access?***
   Based on the gathered 5-tuple data and the compromised state, I recommend the following actions:

- **Isolate the host:** Immediately disconnect the compromised target machine (IP: 209.165.200.235) from the network to stop further data exfiltration.

- **Firewall Blocking:** Block the attacker's source IPs (209.165.201.17 and 192.168.0.11) at the firewall. Additionally, close unnecessary ports, specifically port 6200 (used as a backdoor) and port 21 (FTP).
- **Reset Credentials:** Since the /etc/shadow and /etc/passwd files were compromised, force all users (especially root and analyst) to change their passwords immediately and ensure they comply with the current security policy.